

ADITYA KUMAR

Third Year Undergraduate

Department of Computer Science and Engineering

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology Kanpur	9.84/10
2021	CBSE(XII)	Boston Public School, Agra	97.6%
2019	ICSE(X)	St.Peter's College, Agra	97.2%

Scholastic Achievements

- Emerged as **Institute Winner** in the **Quant Trading Winter School** organised by **IIT Delhi** powered by **Optiver** (2023)
- Received the **Academic Excellence Award** for outstanding and exceptional academic performance at IIT Kanpur (2021-22)
- Secured **All India Rank 184** in **Joint Entrance Examination (JEE) Advanced** among 150,000 candidates (2021)
- Secured **All India Rank 579** in **Joint Entrance Examination (JEE) Mains** among the 1,200,000 candidates (2021)
- Selected for **KVPY Fellowship Award** twice, with an **AIR 149** and an **AIR 670** out of 100,000+ candidates (2019, '20)
- Among the **National top 1%** who qualified the National Standard Examination (**NSE**) in **Chemistry** by IAPT (2020)
- Qualified for the **Indian National Maths Olympiad(INMO)** organised by HBCSE, TIFR along with **123** students (2020)
- Shortlisted to appear for the **Indian National Astronomy Olympiad (INAO)** along with **256** other candidates (2019)

Key Projects

CL Accumulator and Zerocoin Analysis (SURGE, Prof Manindra Agrawal, IIT Kanpur)  (May'23 - Jul'23)

- Studied **CL Dynamic Accumulators**, its implementation using **RSA** and its usage in solving the membership problem
- Analysed the security of **Zerocoin** and showed that it was secure against many double spending and anonymity attacks
- Designed and implemented a **Smart Contract** to share files anonymously on **Hyperledger Fabric Blockchain** using **Go**

What's Next (Course Project, Software Development, Prof. Indranil Saha, IIT Kanpur)  (Jan'23 - Apr'23)

- Developed a web-based application using **NodeJS**, **Express** (backend), **ReactJS** (frontend) and **MongoDB** (database)
- Added the feature that enables Admins (club coordinators) to add regular and special events and get their clash warnings
- Created a dashboard displaying event information, highlighting the special events and enabling users to like and unlike events

DES and AES Decryptor (Course Project, Modern Cryptology, Prof. Manindra Agrawal, IIT Kanpur)  (Jan'23 - Apr'23)

- Explored various cryptographic Algorithms and Techniques including **RSA**, **ECC** and Hashing Algorithm like **SHA256** etc.
- Constructed a **6 round DES** (Data Encryption Standard) decryptor based on the **Differential Cryptanalysis** technique
- Decrypted a simplified version of **5 round AES** (Advanced Encryption Standard) incorporating **Square Attack** technique

Space Sheriff (Internal GameJam, GameDev Club, IIT Kanpur)  (Oct'22 - Nov'22)

- Learned about **Object Oriented Programming** and basics of Game Development with the help of **Unity** Game Engine
- Worked with a team of 5 Secretaries to develop a **2-D** shooting game with 5 levels using our own scripts and UI/UX interface

CF Stress (Programming Club, SnT Council, IIT Kanpur)  (May'22 - Jul'22)

- Analysed working of cfstress.com, a website that generates the smallest test case that a particular C++ program is failing
- Explored various technologies including **Docker**, **MongoDB**, **Mutex**, **Waitgroups**, etc. and implemented them using **Go**
- Implemented the backend of a student registry webpage using MongoDB Database and a simple dictionary using Golang

Technical Skills

- **Programming Languages:** C, C++, C#, $\text{\LaTeX}$, Python, Java, Go, HTML, CSS, Javascript, Verilog, MIPS, Bash
- **Software and Libraries:** Git, Unity, Fusion 360, DaVinci Resolve, MS Office, MongoDB, NodeJS, React, Express

Positions of Responsibility

Secretary (GameDev Club, IIT Kanpur) (Sep'22 - Apr'23)

- Coordinated in the organisation of the **SnT Pavilion** to introduce freshers to the three major areas of game development
- Explored the fundamentals of **Augmented Reality (AR)** and its implementation in contemporary game development
- Debugged games to be hosted on Itch page of the GameDev Club and helped in maintaining the Instagram page of the club

Relevant Courses

(*) - A* grade for outstanding performance

Fundamental of Computing*	Data Structure and Algorithm*	Logic for Computer Science*
Discrete Mathematics	Software Development	Computer Organisation
Linear Algebra and ODE	Modern Cryptology	Probability for Computer Science
Fluid Mechanics and Rate Processes*	Introduction to Economics*	Introduction to Electronics*

Extra-Curricular Activities

- Mentored **20** mentees in the project **Cryptographic Applications of Number Theory** under **Stamatics** IIT Kanpur
- Represented IIT Kanpur in the **Blitzkrieg** online chess competition against **University of New South Wales, Australia**
- Won the second prize in **Be an Angel**, a competition to test investing skill organised by Entrepreneurship Cell IIT Kanpur
- Developed a shooting based 3-D game, **Certain Death** as a part of 5 member team during Fresher's Showdown *Itch Page*
- Won the first place in Fresher's **Anime Music Video (AMV)** making competition organised by the Anisoc IIT Kanpur